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R e s e a r c h P r e s e n t a t i o n a n d P r o p o s a l

Compound Flood Modelling, Bayesian Data Assimilation, Production-scale Continental Modeling

Selected work across four research and applied case studies,
spanning the full hazard – exposure – risk pipeline.

Hydrology

Flood risk modelling

Bayesian + ML methods

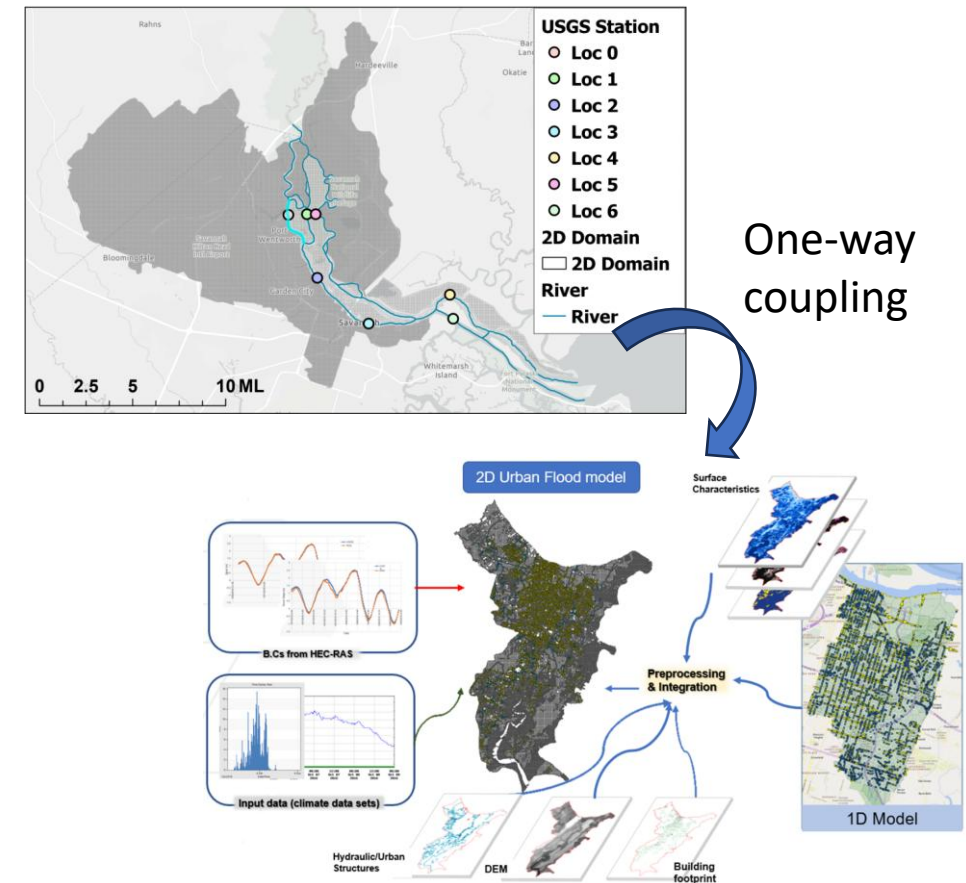
Fluvial, urban flooding

August 2026

Physical Modeling, ML Surrogates, and Remote Sensing: A Compound Flood Case Study

Project Overview

- Goal: Quantify flood drivers interact to produce urban flooding — and determine which driver controls where
- Objectives:
 1. Build and validate a coupled hydrodynamic model capturing surface and drainage-network flow for for 1233 km^2 urban watershed
 2. Quantify driver interactions across a designed scenario ensemble
 3. Verify predicted extent independently against satellite observation
 4. Suggest a methodology to quantify uncertainty and decrease it
- Methods:
 1. Construction of coupled HEC-RAS-PCSWMM flood model for compound urban flood with flexible resolution to save computational cost.
 2. Validated against Hurricanes Matthew, Irma, Dorian, Idalia at five USGS gauges
 3. Latin Hypercube ensemblescenario; PLSR / Random Forest / XGBoost surrogates
 4. Independent verification: Sentinel-1 SAR
 5. Application of Bayesian Data Assimilation (to combine physical model as prior and measurement as likelihood)



WHAT THIS DEMONSTRATES

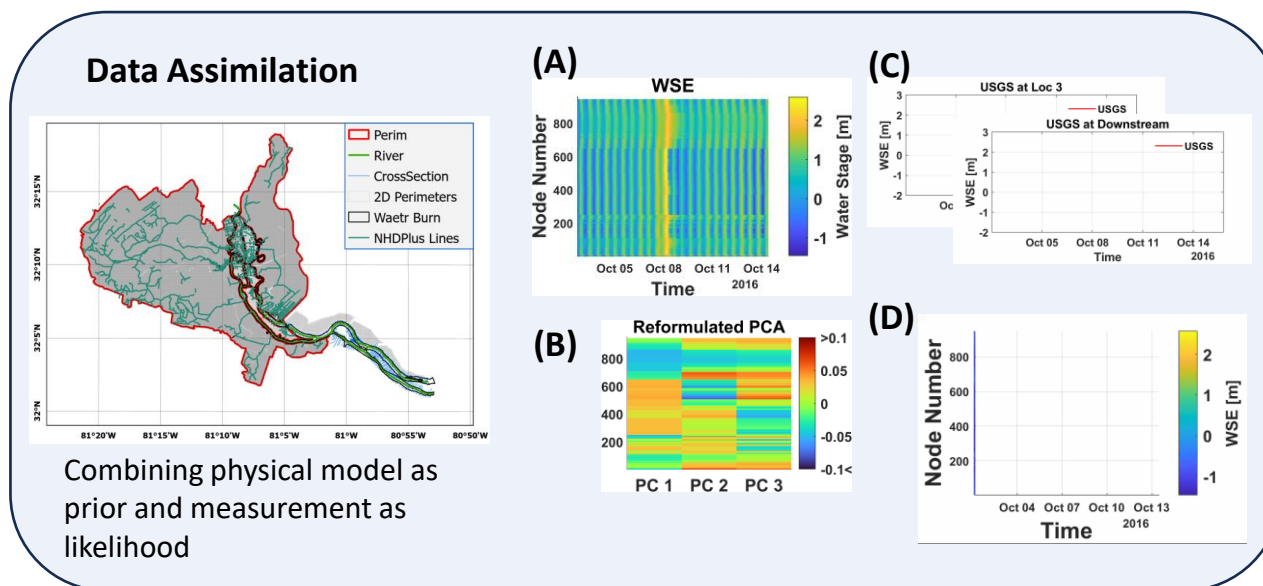
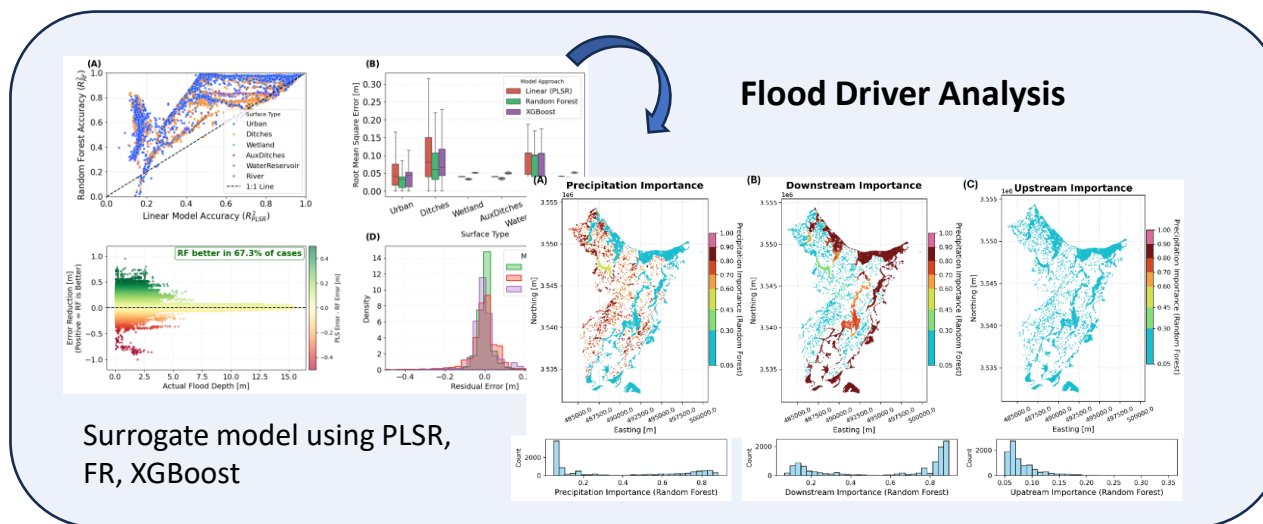
Compound flood modeling, coupled modeling, Bayesian Data Assimilation, ML based flood driver analysis

Section 1: Research closely aligned with CIROH

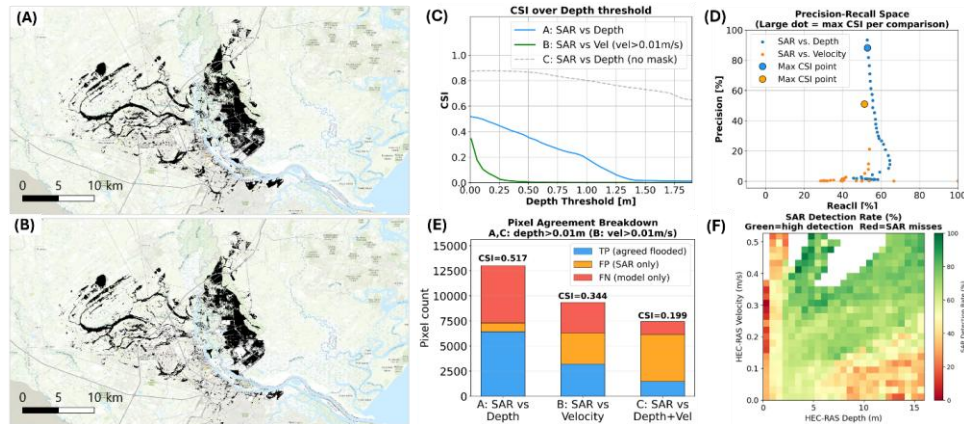
Physical Modeling, ML Surrogates, and Remote Sensing: A Compound Flood Case Study

- Key Results:

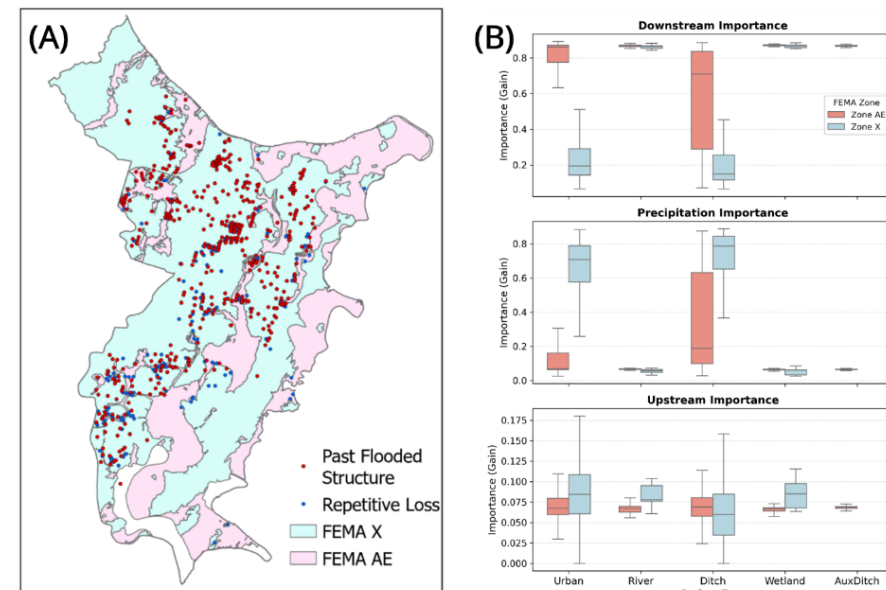
1. Spatially heterogenous response of water depth, residual time to flood drivers (precipitation, upstream condition, precipitation magnitude)
2. Regarding fluvial flood, spatial correlation is maintained through 4 different hurricanes (Matthew, Irma, Dorian, Idalia)
3. FEMA Zone is weighted to one driver (surge), while results show Zone AE is influenced by precipitation than surge.
4. Processed SAR showed a good validation material, suitable for rural areas, not for urban areas with limitations
5. Possibility of an optimization of flood network. Only 3 of 5 stations are necessary, proved by Bayesian data assimilation (DA)



Component	Connection to CIROH Hydrology Modeling
Coupled 2-D hydraulic modeling	Optimal method for operational compound flood modeling
Surrogates for driver attribution	Identification of forecast uncertainty in riverine-coastal system; spatial distribution of driver dominance
Verification with remote sensing data and global data product	Remote-sensing evaluation of model skill, and identification of where the benchmark fails
Bayesian data assimilation	Fusing sparse observations with physical model priors to reconstruct flood state in data- sparse domains
Limited application of FEMA zone in compound flood area	risks in using existing flood-risk datasets for operational assessment



Characteristics of SAR compared to flood model results
(possible failure at extreme condition (too shallow and
to fast) Froude-based assumption



Different dominance of flood drivers to FEMA zone

Section 1: Transpirable knowledge, skills, competencies to CIROH

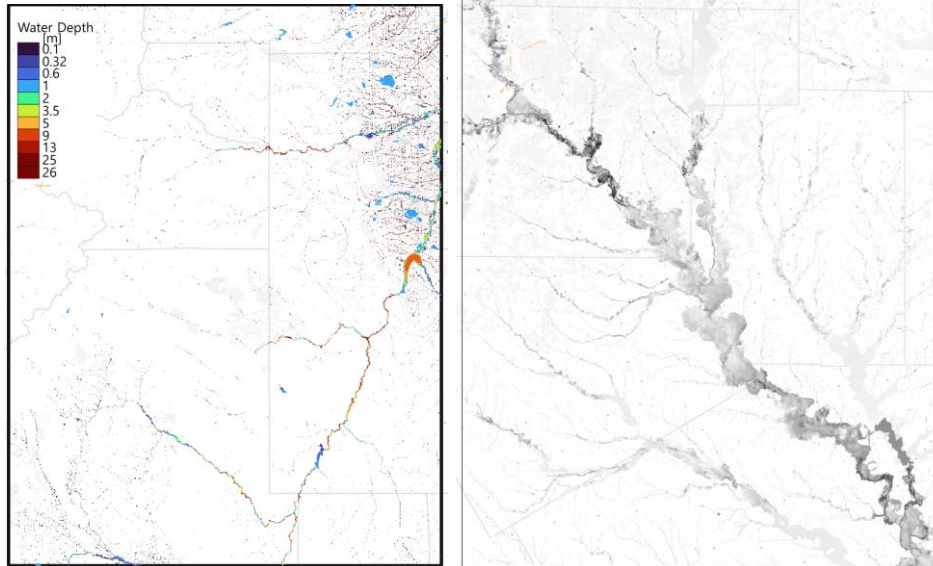
AI/ML — PLSR, Random Forest, XGBoost surrogates on physical model output; tree-based regionalization of return-period discharge at unmeasured boundary points

Large-domain hydrologic modeling — pluvial/fluviol flood modeling, patches of $\sim 75 \times 110$ km ($\sim 66,000$ km²), global operational flood program; MERIT DEM and MERIT Hydro; GloFAS / EFAS as reference

High-performance computing — HEC-RAS on Linux HPC, one core per simulation, AWS-backed data management

Programming and open software development — public repositories for the coupled model, driver analysis, and PIGA; automated HEC-RAS execution with HDF file control

Research studies applying hydrologic models — validated compound flood modeling; PIGA (*J. Hydrology*, 2024); Pipelined model construction



* Large Domain Model example: 1 in 100 yr pluvial flood simulation + 1 in 100 yr fluviol flood simulation for Denver CO (left), Dallas TX (right) of 75 km by 110 km grid

Model setting: grid-based tiling, MERIT DEM, MERIT Hydro
Parameter estimation: tree-based ML (Random Forest, XGBoost)

Solver: HEC-RAS (mesh by Windows GUI) under Linux HPC

AWS for data management, GitHub code repository

Validation: GloFAS and EFAS, Deltras, and FEMA

Understanding of NextGen Water Resource Modeling

A framework, not a model — it composes models rather than computing water balance itself.

- **BMI 2.0:** model-agnostic interface (initialize, update, get_value, set_value, finalize).
The bridge that makes models swappable and couplable.
- **Hydrofabric** (OGC HY_Features): divides, flowpaths, nexus points. The shared spatial addressing scheme.
- **Model formulations:** CFE (conceptual rainfall-runoff), Noah-OWP-Modular (land surface fluxes), TOPMODEL, CASAM, LSTM.
- **Process modules:** PET, SoilMoistureProfiles, SoilFreezeThaw.
- **t-route:** 1-D network routing; includes reservoir modules and streamflow DA against USGS gauges.
- **NGIAB:** containerized deployment; plugin-based model integration without editing the build.
- **NRDS:** community-editable CONUS data stream.
- **Evaluation & calibration:** TEEHR, gval; ngen-cal.

With NextGen / NGIAB



Current and Contributable research topics

Research that has used NextGen

Research Topic	Applied / Recent Study
Data Assimilation Coastal coupling	Ensemble Kalman filter module assimilation of in-situ streamflow into CFE Inclusion of GeoClaw initiated in 2022.
ML surrogate for FIM	HAND-FIM correction via HEC-RAS results: -18% false alarms, +26% CSI (<i>EM&S 194, 2025</i>)

Ideas to advance the framework

Gap	Opportunity
Data Assimilation is gauge only	Possible inclusion of geospatial approach of remote sensing data
BMI-native 2-D hydraulic model Supplement to FIM	Conventional 2D surface routing model could be added; GeoClaw for coastal area, not floodplain FIM is based on geography not physical model, effective post processing. Possible disadvantages under highly dynamic conditions
Addition of urban flood model	A pipe-network formulation (SWMM-class) for pluvial and drainage-controlled flooding

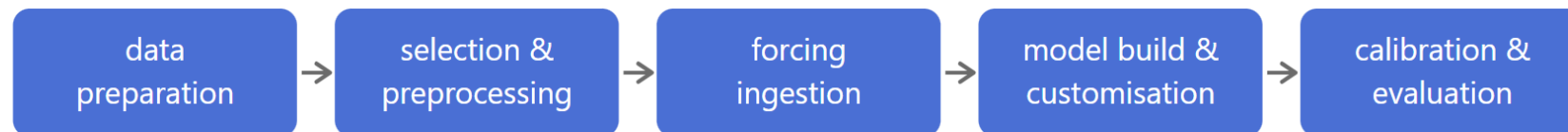
Improve of engagements of researchers to the NextGen Framework

Q: Why Should Researchers Use NextGen?

A: The practical cost of hydrologic modeling isn't the science only— it's also the setup.

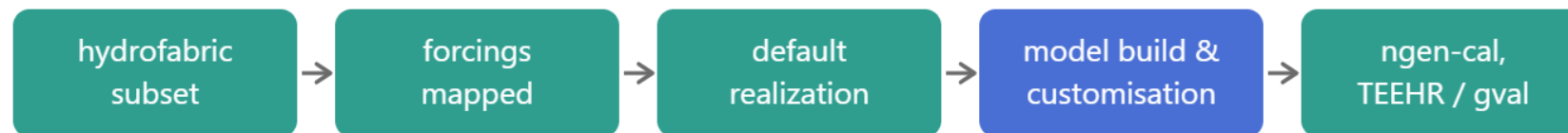
- Save the setup time — no separate effort for input preparation or boundary condition generation
- Start from a working configuration — forcings, hydrofabric, and defaults are already supplied
- A paved path — calibration (ngen-cal), evaluation (TEEHR, gval)
- Low barrier by design — containerized via NGIAB; adding a model needs no build-system work
- One entry point for many discipline — hydrologists, ML researchers, and software developers build on the same foundation

Conventional workflow



■ researcher's own work

With NextGen / NGIAB



■ researcher's own work

■ supplied by the framework

A shared framework enables research questions that no single project can address.

Model combination becomes a tractable research question — different formulations may make different results. So optimized combination of models can be achievable more easily than before.

Process-based and machine learning models compete on equal terms — an LSTM and a conceptual model are both formulations; the framework privileges neither

A continental reference that already runs — NRDS provides a continuously updated CONUS baseline against which any study may be measured.

Model comparisons become meaningful — a common hydrofabric, common forcings, and common evaluation tools enables compare with less uncertainty due to process selection

Research reaches operations — a defined research-to-operations pathway toward the National Water Model

The direct benefit to the contributor.

Continental-scale validation at no cost — a contributed configuration is exercised regularly

Rapid identification of failure modes — many users across many basins surface edge cases that a single-basin study would not

Contributions become infrastructure — work that others build upon rather than read once

Longevity beyond the funding period — a contributed configuration continues running after a project closes

A low-friction submission process — NRDS accepts parameter updates through a documented issue template, not a proposal cycle

A path to operational use — a defined route from contribution toward operational forecasting

Physics-Constrained Machine Learning for Flood Inundation in NextGen

- Motivation: FIM is an efficient post-processing method: NWM discharge is converted to inundation extent through a synthetic rating curve built on the uniform-flow assumption. That assumption holds where flow is channel-confined and fluvially driven. It degrades as floodplains engage and downstream controls activate — conditions that intensify with event magnitude. Moving toward the coast, tidal and surge control introduce backwater the method cannot represent, and compound forcing could be hard to be presented, since discharge is the only input.
- Solution: Add constraints from physical condition such as physics informed machine learning.
- Goal: Determine whether, and if so where low-complexity FIM fails, then correct it with physics-constrained machine learning at a cost that scales nationally.

Phase I — Diagnosis

- Quantify HAND-FIM skill degradation across return period, terrain class, and downstream control
- Establish observability-aware evaluation — condition SAR-based skill on where the benchmark is reliable

Phase II — Correction

- Generate subgrid 2-D ensembles as training data, indexed to the hydrofabrics in NexGen framework
- Train a physics-constrained correction to the synthetic rating curve
- Extend to spatial data assimilation of satellite flood extent

Outline for an advanced research study

Methods

Phase I — Diagnosis

Benchmarks from existing high-fidelity models: ras2fim library, the validated Savannah model

Stratify HAND-FIM error by return period, slope class, confinement, and downstream control type

Sentinel-1 pipeline: thresholding, permanent-water masking, mesh projection, CSI/POD/FAR

Phase II — Correction

Objective: Rating curve to rating surface

- HAND's synthetic rating curve is single-valued: one discharge, one stage. Compound forcing has no entry
- Ensemble discharge and downstream stage a two-variable surface: $\text{stage} = f(Q, \eta \text{ of downstream})$
- Coastal boundary supplied by existing SCHISM total water level
- **Training data generation at scale**
- Subgrid hydraulics cost will be tested
- **Physics-constrained learning**
- Target: 2-D stage vs. rating-curve stage — multiplicative error structure
- Features read directly off the subgrid tables: conveyance ratio (floodplain/channel), wet fraction, ,downstream stage index, reach slope, return period
- Gradient boosting with **monotonicity constraints** on physically-signed predictors
- Leave-one-basin-out cross-validation

Anticipated Result

Phase I

- HAND-FIM error as a function of return period $\times$ terrain class $\times$ downstream control.
- Classification of zones: CONUS into the three tiers: HAND adequate, rating-surface correctable, 2-D required (coastal).
- *Benchmark methods aided with remote sensing (needs more study)*

Phase II

- **A two-variable rating surface**
stage = $f(Q, \eta \text{ of downstream})$ — physically interpretable, transferable to ungauged reaches, consuming NWM's existing SCHISM total water level.
- **Subgrid property library**
Hydrofabric-indexed, contributable through NRDS. The first large-sample labeled dataset for flood inundation.
- **DA applied surrogate subgrid**

Summary of impact and benefit of the research

Operational

FIM coverage is nearly complete under the BIL mandate; fidelity is the next problem. Phase I identifies where fidelity is lacking. Phase II supplies it inside the existing pipeline — a corrected rating relationship, not a replacement system.

Scientific

Resolution and complexity thresholds for large-domain inundation modeling remain unsettled in the literature. This produces a quantitative answer, stratified by terrain and forcing regime.

Community

The subgrid property library is contributable through NRDS and deployable through NCIAB. It can be shared within the framework

Extensible

The same architecture accepts spatial data assimilation, extending NextGen's gauge-based EnKF/PF module to satellite flood extent, possible application of Bayesian data assimilation.

Summary

I am open to research, applied, and regulatory roles.



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L O C A T I O N

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L A T E S T

Water Resource Engineer, WaterWorks for now

Referees and full CV available on request.